Jordan Cao

Computer Science | Boba Enjoyer | 647-383-8878 | jordancao77@gmail.com | LinkedIn | Github

Education

Wilfrid Laurier University — Waterloo, ON
BSc in Computer Science — *2026*

Experience

Software Engineering Intern — Jan 2026 – May 2026
Mercury — *San Francisco, CA (Remote)*

- Designed and implemented **pause/unpause functionality for autotransfer rules** in a large-scale **Haskell** fintech backend, spanning database schema changes, domain types, SQL queries, and API handlers across **140+ commits**.
- Built **database migrations and schema evolution** for autotransfer rule lineage tracking, including backfill scripts, NOT NULL constraint migrations, and a new **pause table with indexing**, ensuring zero-downtime deployments on production banking infrastructure.
- Extended **type-safe Esqueleto SQL queries** and domain model conversions to surface pause status and lineage data through the API, maintaining correctness guarantees via Haskell's type system and comprehensive **HSpec test suites**.
- Contributed to **partner bank rollover logic** by implementing autotransfer disablement during account migrations, coordinating across multiple service boundaries in a **Nix-based monorepo** with **Temporal workflows**.

Research Assistant - Machine Learning & Evolutionary Algorithms — Sep 2025 – January 2026
Wilfrid Laurier University — *Waterloo, Kitchener*

- **Utilized Phikon-2 on the Whole Slide Imaging (WSI) dataset** to extract **50,000+ high-dimensional features per slide**, enabling **faster downstream classification and more compact medical image representations**.
- **Optimized CNN architectures** through iterative experimentation with evolutionary strategies, resulting in **a 20% reduction in redundant features** and **shorter training times without sacrificing accuracy**.

Data Specialist — Jan 2025 - April 2025
Wilfrid Laurier University — *Waterloo, Kitchener*

- Leveraged **Python and ML libraries** (**NumPy, Scikit-learn, PyTorch, Matplotlib**) for predictive analytics and donor segmentation, identifying **high-propensity prospects with ∼85% accuracy** through advanced modeling and data mining.
- Created **interactive dashboards and automated reporting tools** for stakeholders using visualization libraries and structured SQL queries to support real-time analytics.
- Validated and cleaned **100,000+ alumni records**, ensuring data accuracy, consistency, and completeness by removing duplicates and resolving inconsistencies, which improved data reliability for reporting.

Projects

Donor Prediction & Segmentation Dashboard | *Python, SQL, VBA,*

- Designed an end-to-end **predictive analytics workflow** using **Python, SQL, and visualization tools**.
- Built visualizations in **Matplotlib and Power BI** to communicate findings to non-technical stakeholders.
- Provided actionable insights that improved donor outreach strategies.

Skills

Languages: Haskell, Java, Python, C/C++, SQL, JavaScript, TypeScript, HTML/CSS, VBA, Assembly
Developer Tools/Specialized Skills: Git, Nix, Docker, PostgreSQL, React, Node.js, AI/ML, CNNs, Image Classification, Data Processing, Eclipse, Supabase, AWS, Power BI, Temporal, GHCi, Cursor, Claude Code
Libraries: Esqueleto, Persistent, HSpec, Pandas, NumPy, Matplotlib, PyTorch, Scikit-learn, TensorFlow